A Multimodal Feature Fusion Network for Building Extraction With Very High-Resolution Remote Sensing Image and LiDAR Data

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***Abstract*— Building extraction from remote sensing images is extremely important for urban planning, land-cover change analysis, disaster monitoring, and so on. With the growing diversity in building features, shape, and texture, coupled with frequent occurrences of shadowing and occlusion, the use of high-resolution remote sensing image (HRI) alone has limitations in building extraction. Therefore, feature fusion using multisource data has gradually become one of the most popular. However, the unique characteristics and noise issues make it difficult to achieve effective fusion and utilization. Thus, it is very challenging to realize the full fusion of multisource data to achieve complementary advantages. In this article, we propose an end-to-end multimodal feature fusion building extraction network based on segformer, which utilizes the fusion of HRI and light detection and ranging (LiDAR) data to realize the building extraction. First, we utilize the segformer encoder to break through the limitations of the traditional convolutional neural network (CNN) with the restricted receptive field, so as to achieve effective feature extraction of complex buildings. In addition, we propose a cross-modal feature fusion (CMFF) method utilizing the self-attention mechanism to ensure the fusion of multisource data. In the decoder part, we propose a multiscale upsampling decoder (MSUD) strategy to achieve a full fusion of multilevel features. As demonstrated by experiments on three datasets, our model shows better performance than several multisource building extraction and semantic segmentation models. The intersection over union (IoU) for buildings on the three datasets reaches 91.80%, 93.03%, and 84.59%. Subsequent ablation experiments further validate the effectiveness of each strategy.**

***Index Terms*— Building extraction, multisource fusion, segformer network.**

I. Introduction

BUILDING extraction has always been one of the hot spots in the field of remote sensing. Accurately extracting buildings from remote sensing images is of great significance, as it plays a crucial role in numerous fields, including urban planning [1], 3D modeling [2], disaster monitoring [3], [4], [5], [6], map services [7], [8], and so on.

With the rapid development of satellite sensor technology, a variety of different sensor data, such as high-resolution remote sensing images (HRIs), hyperspectral images, and light detection and ranging (LiDAR) point cloud data, have provided a solid data foundation for reliable large-area building extraction. At present, most researchers have used only optical images [9], [10] or LiDAR data [11], [12], [13]. However, as the number of buildings increases, the textures, shapes, and other features of buildings tend to diversify. In addition, problems of shadowing and occlusion often occur due to the different acquisition times and angles of remote sensing images [14]. A variety of factors further increase the difficulty of building extraction [15]. Thus, a single data source is no longer sufficient for the need of accurate building extraction. There are obvious drawbacks for single data source. For example, high-resolution images are rich in detail, but they can not solve the problems of shadow and occlusion. They are easily confused by the spectral characteristics of similar features. LiDAR data can avoid the problem of shadow and can distinguish buildings from other features of similar construction materials, such as roads and squares. However, the boundaries of buildings are blurred in LiDAR data. Therefore, more and more scholars are trying to utilize multisource data to improve extraction accuracy.

Multisource data fusion can compensate for the problems existing in single data to achieve improved interpretation accuracy. Nonetheless, without the implementation of suitable methods for utilization, there is a risk of inadequate feature extraction, potentially giving rise to noise interactions among the data, subsequently impacting the results [16]. Consequently, effectively harnessing multisource data for building extraction remains a formidable challenge. Traditional multisource building extraction methods generally utilized simple features of morphology such as color, texture, and geometric information [17], [18], [19], [20] and then followed the rules of early theorems before using machine learning algorithms [21], [22], [23], [24], [25]. However, these methods relied more on manually designed low or mid level image features to identify buildings, lacking the use of high-level features and with very limited extraction accuracy. Afterward, deep learning [26] relied on adaptive learning to achieve feature extraction and fusion, fully mining the multidimensional features of data.

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It shined in the field of multimodal fusion and was widely used for extracting multisource buildings. Zhang et al. [27] classified multisource data fusion into early fusion, mid fusion, and late fusion according to the location of data fusion and the degree of data abstraction. Early fusion is a direct pixel-level fusion to preserve the original features of the image as much as possible. It is mostly used for early multisource fusion because it is simpler, more straightforward, and very easy to operate [28], [29], [30]. Later, Audebert et al. [31] experimentally demonstrated that the early fusion strategy was vulnerable to missing data and noise, and further proposed a late fusion strategy based on residual correction to further recover misclassified pixels. Late fusion generally starts with some simple classification of the earlier data, and the initial classification results are further fused according to a predefined decision strategy to obtain the final classification results. However, late fusion relies too much on the earlier classification results and decision strategy, so it is generally less commonly used. Instead, flexible and versatile mid fusion strategies are preferred by most scholars.

The mid fusion strategy begins by extracting pertinent features from diverse data sources, followed by processing these features to derive conclusive classification results. The feature extraction process is flexible, as the original data can be abstracted at different scales through a hierarchical model and then integrated by assigning appropriate weights. For example, researchers such as Zhang et al. [32], Sun et al. [33], and Yuan et al. [34] have employed two independent multilayer encoder structures to extract multilevel features from input images, thereby capturing more contextual information. Then, the features of the same scale are weighted and fused. In the decoder phase, effective fusion of multiscale features is achieved through a unique design, ensuring the accuracy of remote sensing image interpretation.

Although these methods have obtained excellent results, they still have some unsolved problems. First, they lack attention to the encoder and utilize the traditional convolutional neural network (CNN), which makes the receptive field limited and, thus, consequently makes them unable to utilize the dependency of global and contextual information for feature extraction. This is very unfavorable for the extraction of buildings. Because the shape of buildings tends to be diverse, and the lack of global contextual information will lead to small-sized buildings being ignored after downsampling, some super-large buildings also will not be accurately extracted due to the limited receptive field. Although some scholars [34], [35] tried to make improvements, the effect was not significant. On the other hand, the lack of sufficient consideration for complex buildings and the insufficient utilization of multilevel features will lead to incomplete recovery of the original features of buildings in the decoder part. In addition, for multisource remote sensing data, the existing methods only perform simple superposition or simple weighted fusion without considering the global relationship between features. Thus, the features of multisource data cannot be fully utilized. That is extremely easy to cause the increase of noise to affect the final results.

Therefore, we propose a multimodal feature fusion building extraction model based on segformer [36]. Segformer combines local and global attention to present powerful representations, which has been very fruitful in the field of semantic segmentation. In the decoder part, we consider deeply the characteristics of the buildings. We introduce a multiscale feature fusion module that combines high-level semantic information with low-level semantic details, ensuring the effective fusion of intricate building features across diverse sizes and types. In addition, in order to ensure the full fusion of multisource data and realize the complementary advantages, we design an adaptive multisource fusion module based on the self-attention mechanism, which guides the two different modal data to be fused adaptively. The main contributions of this article are given as follows.

1) We proposed a multisource data fusion network framework based on segformer for end-to-end extraction of buildings, which solves the problem that traditional deep learning networks have a limited perceptual field and consequently underutilize global contextual information.
2) A cross-modal feature fusion (CMFF) module is proposed to ensure the full use of multisource data by guiding the fusion of multimodal data from global contextual information using cross-attention.
3) A multiscale upsampling decoder (MSUD) is proposed for the full fusion of different scales of data in the decoder part. It ensures the effective use of classification information in high-level features while retaining the incorporation of detailed features in low-level features, thus enabling the effective extraction of complex building types.

II. Related Work

A. Multisource Building Extraction Based on CNN

In light of the significant advancements in deep learning within the field of remote sensing, more and more researchers are turning to using CNN for multisource building extraction. Compared to traditional methods, CNN has the capacity to encompass various forms of information, including texture, spectral data, spatial context, and interactions among geographic features. Audebert et al. [31] were the pioneers in utilizing the Segnet network for merging HRI and digital surface model (DSM). Their experimental results proved the advantages of multisource data for building extraction. However, they only performed a simple pixel summation, which resulted in severe negative impacts on the final results when dealing with LiDAR anomalies, such as missing or outlier points. In 2020, Zhang et al. [37] proposed a hybrid attention-aware fusion network (HAFNet), which employed a three-branch fully convolutional network (FCN) to handle diverse data sources and introduced an attention-aware multimodal fusion block to address cross-modal fusion issues. However, the remarkable achievements of HAFNet came at the cost of a large number of parameters. Consequently, Ferrari et al. [38] improved the model and introduced the HAFNet with efficientNet integration (HAFNetE), utilizing EfficientNet-B0 as the core network architecture and further enhancing the multisource data fusion while compressing the original network parameters. The focus of these scholars is on the selection of the encoder and the design of the

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fusion module. Hosseinpour et al. [39] took an additional step by considering the importance of the decoder in building extraction, resulting in the introduction of the cross-modal gated fusion network (CMGFNet). They utilized the designed gated fusion module (GFM) and the residual-like depth-wise separable convolution (R-DSC) to further improve building extraction accuracy.

The above CNN methods have been synthesized from different aspects to promote the development of multisource building extraction. Nonetheless, due to the limitations imposed by CNN convolution operations, achieving significant breakthroughs has become increasingly challenging. As a result, more researchers are shifting their attention toward transformer methods, leveraging new frameworks to enhance interpretational accuracy while building upon the strengths of CNN.

B. Building Extraction Based on Transformer

In recent years, the transformer has made great progress in semantic segmentation, and some transformer models have obtained better performance than CNN in certain contexts. Models such as segmentation transformer (SETR) [40], SegFormer [36], and Swin-UNet [41] have exceled in semantic segmentation by modeling global contextual information through multilayer models. Consequently, many researchers have begun applying transformers to building extraction. They have explored various Transformer variants, including vision transformer (ViT) [42], Swin Transformer [43], and hierarchical transformer network (MiTNet) [44], with experimental results confirming the superior performance of transformer compared to CNN. Furthermore, these results emphasize the importance of hierarchical transformer structures in building extraction. However, these experiments have primarily focused on single high-resolution images and have not leveraged the transformer structure for the fusion of multisource data.

As the transformer has shown exceptional performance in single-source remote sensing, many researchers are now experimenting with their application in multisource remote sensing. Ma et al. [45] introduced crossmodal multiscale fusion network (CMFNet), which uses a crossmodal multiscale transformer (CMTrans) and a multiscale context augmented transformer (MCATrans) to fuse multiscale feature maps of HRI and DSM data, demonstrating outstanding performance. Inspired by Unetformer, Kieu et al. [46] utilized 3D convolution to encode local information while simultaneously learning cross-modal features. They improved the PerceiverIO multimodal network, creating a versatile multimodal remote sensing model that outperformed architectures, such as UNetFormer and SwinUNet. Zhang et al. [47] introduced the multisource image fusion transformer (MsIFT), which combined local features and global attention mechanisms by using a combination of CNN and transformer to enhance the model's robustness against spatial disparities in multisource data. Many other outstanding researchers have also investigated multisource remote sensing transformers, demonstrating the advantages of transformers over CNN in multisource data applications.

Inspired by these excellent works, we have adopted the Segformer encoder as the primary encoder to provide global context information and applied transformer to the field of multisource remote sensing building extraction. This ensures the extraction of multisource data features, addresses the limitations of CNN, and enhances segmentation accuracy.

III. Methodology

The main framework in this article is shown in Fig. 1. We utilize two separate encoder networks to extract features from both the HRI and the processed LiDAR elevation data. Both extraction branches are pretrained segformer-b4 networks. Then, the extracted multiscale features of the two-source data are fed into the adaptive feature fusion part named CMFF. The decoder is divided into two main parts. The bottom decoder uses a traditional residual decoder (RD) for upsampling the highest-level semantic features. Each layer then uses an MSUD to fuse multiscale features from different layers, fusing low-level refined semantic features with high-level abstract semantic features to ensure that the multiscale features are fully utilized.

A. Encoder Network Framework

We use the segformer network as the backbone of the encoder. It is a hierarchically structured transformer encoder capable of extracting both deep high-level semantic features and refined low-level features. Segformer uses a self-attention mechanism at different levels through a hierarchical structure that allows the model to focus on features at different scales and integrate information from all levels. This allows the segformer to obtain both local and global contextual information, thus improving its segmentation performance. The extraction of buildings is challenging due to their varied shapes, complex building materials, fluctuating heights, and the dense distribution of buildings in some areas. In addition, another challenge in building extraction is the accuracy and continuity of edge extraction, which can easily lead to confusion and blurring of edges. Traditional CNN uses convolutional operations to extract local features and progressively extract global features through pooling operations, thus lacking attention to global contextual information. Segformer uses a unique self-attention mechanism that can calculate the relative importance of each location in relation to other locations. This allows the model to better control the perceptual field and better capture information at different scales in the image. Consequently, the model can more effectively correlate different features, ensuring robust feature extraction, especially in complex building areas. Segformer also improves the self-attention module by proposing efficient self-attention, which reduces the amount of network computation while retaining the advantage of self-attention. The self-attention mechanism in segformer significantly elevates its perceptual field compared to CNN. It produces both local and nonlocal attentional features, effectively establishing connections between semantic information and context. This proves highly advantageous for both global and local feature extraction. In summary, the segformer network boasts significant advantages in the realm of building extraction, making it exceptionally well-suited for this specific task.

The segformer framework is divided into four layers. The first layer uses the convolution kernel for 7×7 convolution and a step size of 4, and padding is 3. Choosing a 7×7 convolution kernel here can more effectively capture larger local spatial information compared to smaller convolution kernels,

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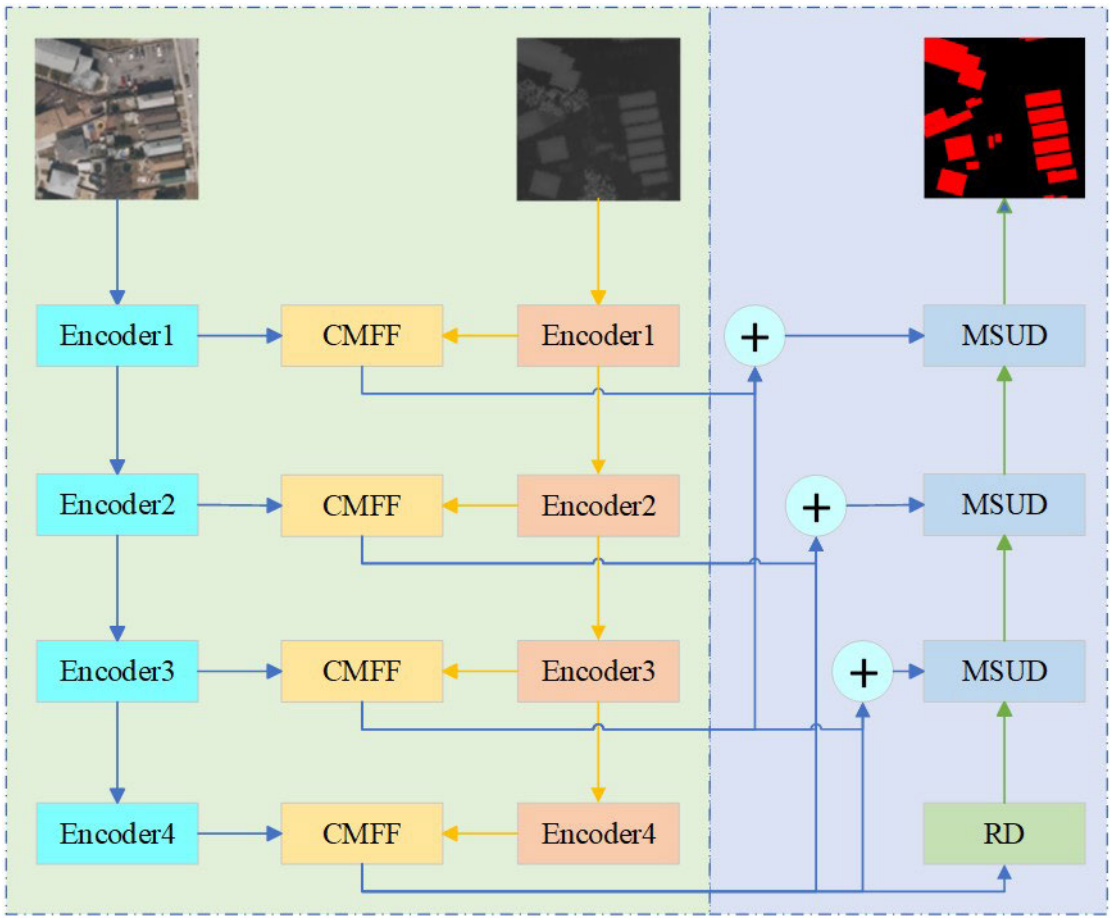


Fig. 1. Illustration of our architecture.

effectively extracting the primary features of the input image, such as edges and textures, thus preserving more spatial details when processing the image. This is important for subsequent feature extraction and segmentation tasks. We use H and W to represent the height and width of the input data, and C represents the number of channels. Therefore, the input size can be described as $H \times W \times C$. After the first layer, the image size becomes $(H/4) \times (W/4) \times C_1$, each layer of convolution kernel for 3×3, and a step size of 2, and padding is 1. This is equivalent to shrinking the original image to half. We extract the multiscale features from HRI and LiDAR data by two independent segformer encoder structures and then input the extracted features to the fusion section for the next step.

B. Cross-Modal Feature Fusion

We present an adaptive multimodal data fusion module based on the self-attention mechanism that fully considers every position of the multisource features to ensure full utilization. For multisource data, the challenge lies in preserving distinctive features from each data source to achieve comprehensive fusion while mitigating the impact of noise from one source on the others. The conventional fusion method treats HRI and LiDAR data as equal contributors, a practice that is evidently not justifiable. In general, HRI data tend to contain more important data features, while LiDAR data play a secondary and complementary role. Therefore, we need a more reasonable fusion method to ensure the full use of both of them to achieve complementary advantages. As shown in Fig. 2, our fusion method is divided into two phases. In the first stage, we use the cross-attention mechanism to dynamically capture the correlation between the two modalities. Specifically, the input high-resolution image is first mapped twice differently, and the LiDAR data are mapped once linearly. As shown in (1) and (2), Q_i^L is the mapping result of the ith layer LiDAR data sum, W is the trainable weight, H_i represents the ith layer high-resolution image features, and L_i represents the ith layer LiDAR data features. The mapped features Q and K are then inner producted to obtain the attention matrix, and the weights are obtained by softmax. Later, the obtained weights are used to aggregate the features. After the aggregation is completed, the features are fused with the original features to compensate for the loss of semantic information. By using cross-attention, each position of the feature map is weighted and fused by using the global information. This process can be specifically described as follows:

$$Q_i^L = L_i W_{Q_i}, \quad K_i^H = H_i W_{K_i}, \quad V_i^H = H_i W_{V_i} \tag{1}$$

$$F_i = \text{Softmax}\left(\frac{Q_i^L {K_i}^{H^T}}{\sqrt{d_{\text{head}}}}\right) {V_i}^H. \tag{2}$$

In the second stage, a residual structure is added to enhance the generalization ability of the model, effectively reducing the gradient disappearance and gradient explosion problems.

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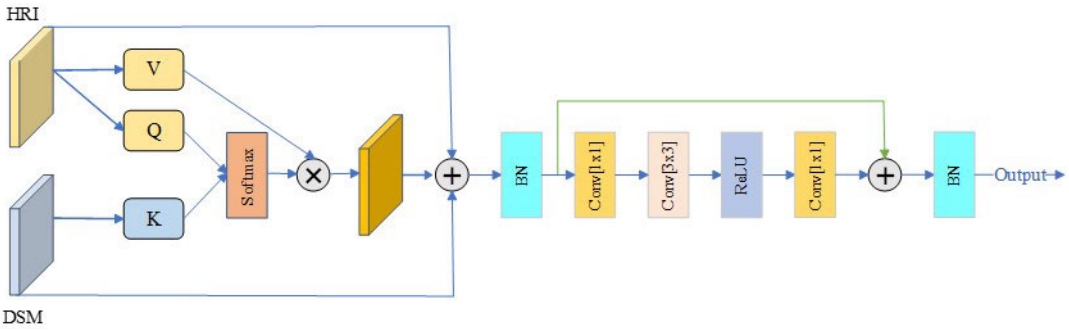


Fig. 2. Structure of the CMFF module.

It is estimated as:

$$\text{Sum}_i = F_i + H_i + L_i \tag{3}$$

$$\text{Out} = \text{Conv}_{1\times1}(\text{ReLU}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(\text{Sum}_i)))). \tag{4}$$

$\text{Conv}_{1\times1}$ represents a convolution operation with a convolution kernel of 1×1.

C. Multiscale Upsampling Decoder

For most decoders, features at different scales are stitched together directly using jump joints, and the original resolution is gradually restored. Such an approach obviously has certain disadvantages, as it tends to cause information redundancy and does not fully exploit the unique roles of high-level features and low-level features. In general, high-level semantic features have more obvious classification information and can guide low-level semantic information well. Therefore, we propose a multiscale upsampling decoder (MSUD). The fusion of multiscale features is achieved by generating weights using global information through the unique classification information of high-level features. This approach preserves both the abstract classification information from high-level features and the detailed contours from low-level features, maximizing the strengths of each source. The detailed process is shown in Fig. 3, where the low-resolution features are first recovered by bilinear interpolation upsampling. The output of the previous decoder is used to obtain the maximum perceptual field and global context information by global average pooling. Then, the corresponding weights are calculated using the sigmoid function and multiplied with the fused features in this layer to achieve the fusion of the results. Ultimately, the results of the upper layer features are summed to further fuse the high-level semantic features with the low-level semantic features. Furthermore, prior to the fusion step, following the approach outlined in [39], we employ summation to directly fusion features from the first two layers with those in the current layer through upsampling. These combined features are subsequently fed into the decoder module as the representative features for the current layer. We use F_i to represent the fused features; Out_{i-1} denotes the output of the previous decoder layer, Up stands for the upsampling operation, Mul indicates the multiplication operation, and AAP stands for AdaptiveAvgPool. The entire decoder can, thus, be described as:

$$F_i = \text{Mul}(\text{Sig}(\text{AAP}(\text{Up}(\text{Out}_{i-1}))), \text{Up}(F_i)) \tag{5}$$

$$\text{Out}_i = \text{Bn}(F_i + \text{Out}_{i-1}). \tag{6}$$

D. Loss Function

The proposed loss functions for this study are the binary cross-entropy (BCE) loss and the dice loss; their formulas are shown in (7) and (8). The final loss function is shown in (9), where p represents the actual label value and $\hat{p}$ represents the model prediction value. The BCE loss is one of the most commonly used loss functions for dichotomous classification and has shown very good results for dichotomous classification. In addition, the ratio of the number of building types to the number of backgrounds in all three datasets used is close to 1:3, so the dice loss is used to mitigate the negative effects of sample imbalance. The two loss functions work together to show excellent extraction results

$$L\left(p, \hat{p}\right) = -\left(p \log\left(\hat{p}\right) + (1-p)\log\left(1-\hat{p}\right)\right) \tag{7}$$

$$L_{\text{dice}} = 1 - \frac{2|p \cap \hat{p}| + \xi}{|p| + |\hat{p}| + \xi} \tag{8}$$

$$\text{Loss}\left(p, \hat{p}\right) = L_{\text{bce}}\left(p, \hat{p}\right) + L_{\text{dice}}\left(p, \hat{p}\right). \tag{9}$$

IV. Experimental Results and Analysis

A. Evaluation Metrics

There are many evaluation criteria commonly used in the field of remote sensing. Since the special characteristics of buildings lead to a large proportion of background elements, the accuracy cannot determine the extraction effect of the model to a large extent, so we decided to use the intersection over union (IoU) of buildings as the main evaluation criterion. The IoU is the intersection of the true and predicted values of a pixel divided by the concatenation of the true and predicted values of the pixel, which gives a good indication of how well the building is extracted. In addition, the final evaluation of the model is made through accuracy and $F1$-score as secondary evaluation criteria. They are calculated as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

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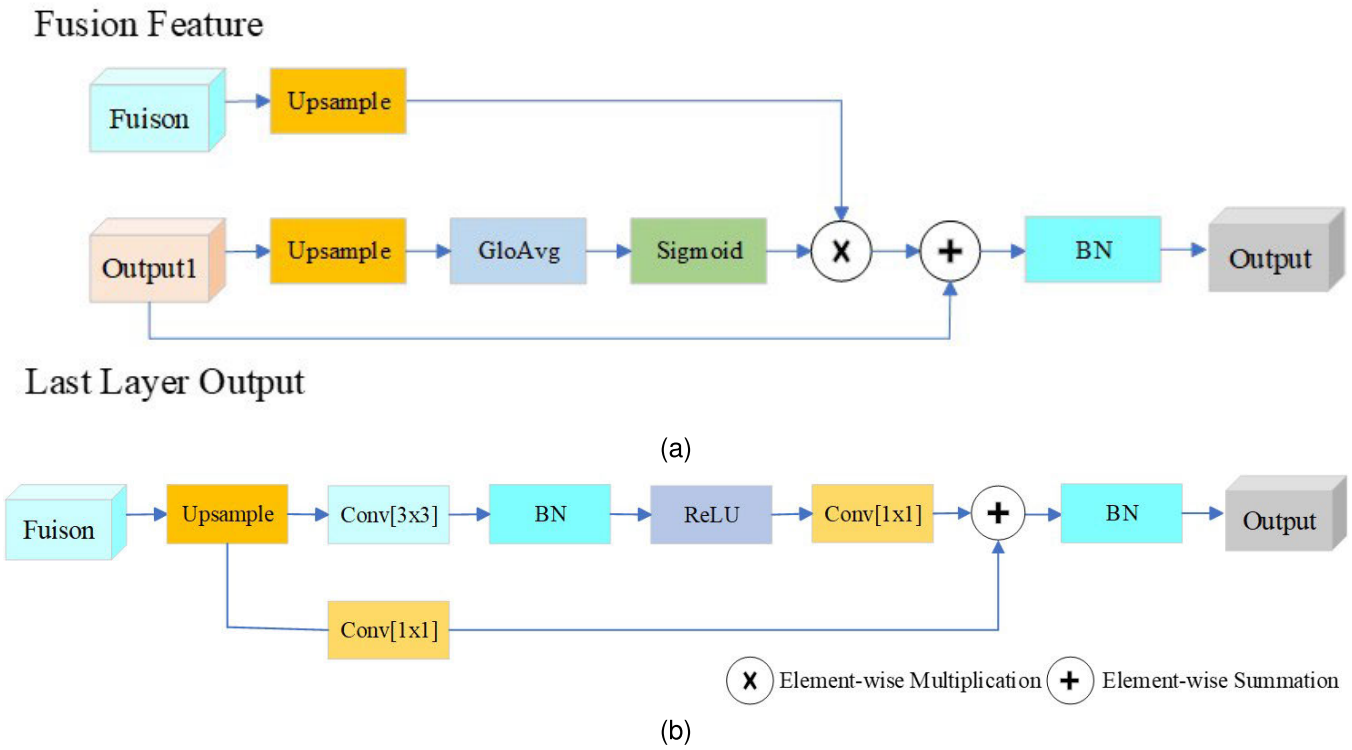


Fig. 3. Structure of (a) MSUD module and (b) RD module.

$$F1-\text{score}=\frac{2*\text{Precision}*\text{Recall}}{\text{Precision}+\text{Recall}} \tag{13}$$

$$\text{IoU}=\frac{\text{traget}\cap\text{prediction}}{\text{traget}\cup\text{prediction}} \tag{14}$$

where TP is the true positive, FP is the false positive, TN is the true negative, and FN is the false negative.

B. Implementation Details

In this experiment, the original dataset was first cropped to a size of 512 × 512 without overlap, and then, the training and test sets were randomly selected in a ratio of 7:3. The parameters were fixed using seeds in order to preserve the experimental parameters for easy replication. The experiments were conducted using the popular PyTorch framework with an RTX 3090 24-GB RAM single-card GPU for training. Some data augmentation was performed on the experimental data, mainly random rotations of the original data, and so on. The optimizer used for the experiments was AdamW [48]. The initial learning rate was configured at 1.5×10^{-4}, and the learning rate was dynamically updated using the ReduceLROnPlateau function that comes with the PyTorch framework.

C. Experimental Datasets

Our experiment utilizes three datasets. Two of the datasets are the Vaihingen and Potsdam datasets. They are provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) and most commonly used in remote sensing. The other one is named USGS. It is provided by the United States Geological Survey (USGS). They all have high-resolution remote sensing orthophotos and LiDAR data in different formats. Fig. 4 shows a sample presentation of the three datasets.

1) Vaihingen and Potsdam Datasets: The Vaihingen and Potsdam datasets are two very well-known 2-D semantic tagging datasets, both collected by airborne sensors. The Vaihingen dataset is a small village in Germany and, therefore, has a large number of individual buildings and small buildings; the Potsdam dataset is a typical historical city with a continuous and large building stock. Both datasets are manually labeled. The Vaihingen dataset consists of 33 images of different sizes, each with a spatial resolution of 9 cm. The image selection process avoids the occurrence of missing data. The orthophotos consist of three bands: NIR, red, and green. LiDAR data are a single-band nDSM. The Potsdam dataset consists of 38 images, each of size 6000 × 6000, with a spatial resolution of 5 cm. Orthophotos are composed of three bands: red, green, and blue; LiDAR data are the same as Vaihingen. The two datasets of HRI data and nDSM data are used as input processing of the data in the experiment.

2) USGS Dataset: The dataset is provided by the U.S. National Geological Survey and consists of 25 high-resolution aerial orthophotos of nine cities and suburbs in the United States, with resolutions ranging from 0.15 to 0.3 m. Each image contains three bands: red, green, and blue; some images contain near-infrared bands. In order to unify the processing, only the red, green, and blue bands are used in our experiment. However, as the LiDAR point cloud data for some of the cities are no longer available for download, a total of 16 images from six cities are selected. The three high-resolution orthophoto

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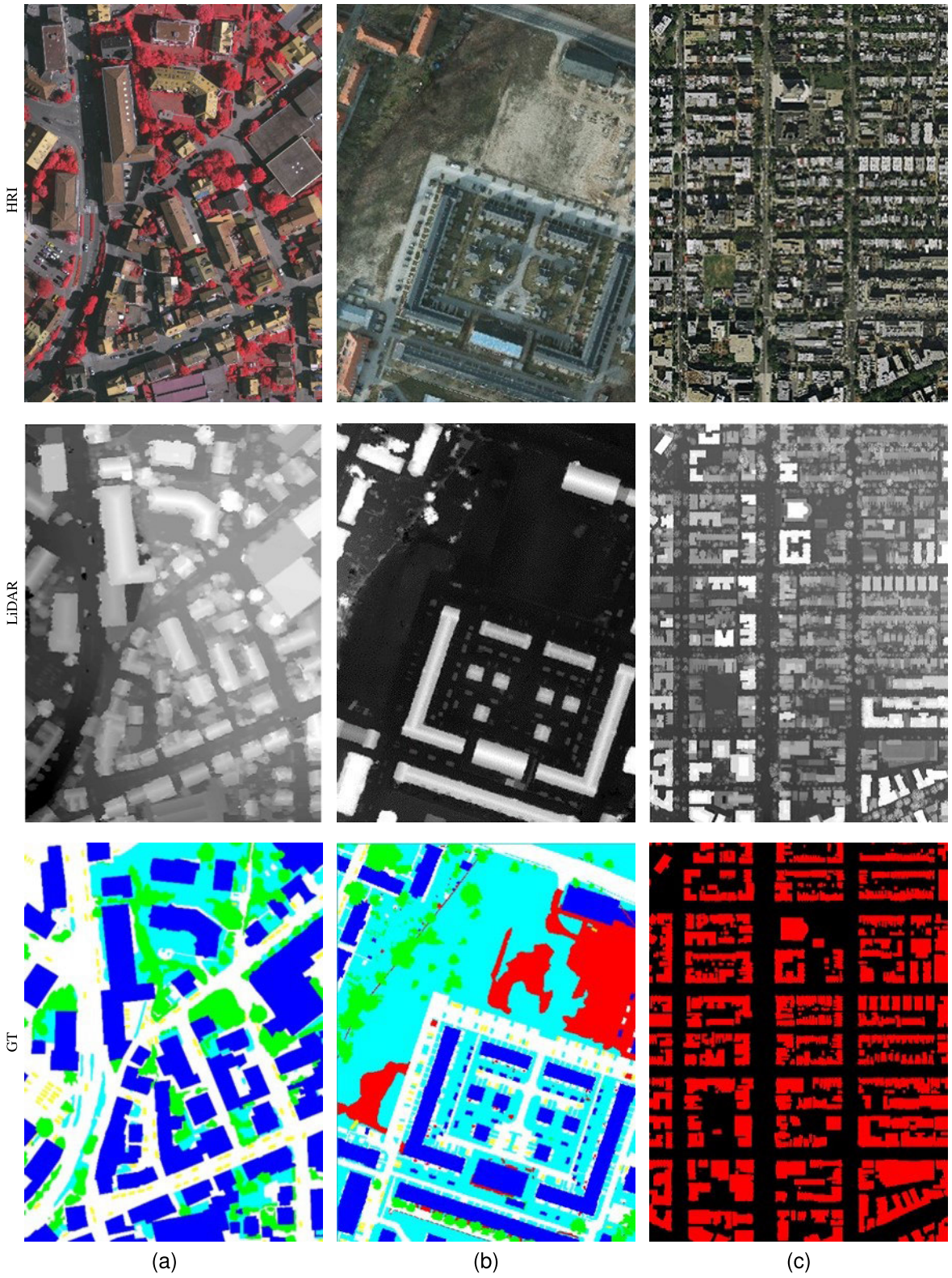


Fig. 4. Examples for the datasets, the corresponding LiDAR, and the ground truth. (a) Vaihingen. (b) Potsdam. (c) USGS.

images of Atlanta are 4000 × 6000, while the rest of the cities are 5000 × 5000. The DSM elevation data are generated with the same resolution as the orthophoto by the ArcMap tool. The image labels are obtained from the Open Street Map (OSM), but each image is manually corrected due to errors in the coordinate system and the timing of image and label acquisition. The dataset contains several representative places, such as urban areas and suburban areas with complex topographical conditions, which is ideal for testing the extraction performance of the model.

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TABLE I
VAIHINGEN EXPERIMENTAL RESULTS FOR DIFFERENT MODELS

Model-Name	ACC	F1-score	IoU
CF-Net	96.63	93.91	88.51
HAFNetE	97.16	94.81	90.14
MultiModNet	97.35	95.20	90.81
CMFNet	96.74	94.10	88.85
CMGFNet	97.26	95.04	90.54
MGCNet	97.06	94.70	89.92
SFAF-MA	97.04	94.65	89.84
CMNeXt	97.34	95.18	90.80
Ours	**97.64**	**95.72**	**91.80**

TABLE II
POTSDAM EXPERIMENTAL RESULTS FOR DIFFERENT MODELS

Model-Name	ACC	F1-score	IoU
CF-Net	97.42	94.94	90.37
HAFNetE	97.86	95.78	91.90
MultiModNet	97.88	95.82	91.97
CMFNet	97.58	95.23	90.89
CMGFNet	97.76	95.60	91.56
MGCNet	97.62	95.32	91.06
SFAF-MA	97.83	95.76	91.86
CMNeXt	97.87	95.83	91.99
Ours	**98.16**	**96.39**	**93.04**

TABLE III
USGS EXPERIMENTAL RESULTS FOR DIFFERENT MODELS

Model-Name	ACC	F1-score	IoU
CF-Net	95.13	86.02	75.46
HAFNetE	96.40	89.86	81.59
MultiModNet	96.49	90.18	82.17
CMFNet	96.45	90.01	81.82
CMGFNet	96.40	89.83	81.53
MGCNet	96.42	89.87	81.60
SFAF-MA	96.66	90.64	82.88
CMNeXt	96.72	90.69	82.96
Ours	**97.03**	**91.65**	**84.59**

D. Experimental Results

1) Comparison With Other Multisource Networks: In order to further validate the performance of our model, we choose several state-of-the-art multisource building and semantic segmentation approaches, including cross fusion net (CF-Net) [49], HAFNetE, multi-modality network (MuiltiModNet) [50], CMFNet, CMGFNet, multilevel gated collaborative network (MGCNet) [51], spatial feature aggregation and fusion with modality adaptation (SFAF-MA) [52], and cross-modal segmentation model for RGB-X (CMXNeXt) [53]. Among them, CF-Net and CMFNet are multisource semantic segmentation models for remote sensing; HAFNetE and CMGFNet are multisource building extraction models; MuiltiModNet is a multisource land classification model for remote sensing; MGCNet and SFAF-MA are multisource semantic segmentation models for other fields; and CMXNeXt is a generic multisource semantic segmentation model for various fields. Tables I–III show the comparison of different models. The quantitative results reveal the exceptional extraction performance of our model across all datasets. This effect is especially obvious in the USGS dataset, which reaches about 1.5%. USGS is more complex and contains many types of buildings in different cities with different resolutions. Thus, it is more effective in verifying the model performance.

Figs. 5–7 show the qualitative results of several experiments. By analyzing the results, our experimental method is more sensitive to the boundaries of the buildings. Our model can identify the boundary areas of the buildings more accurately even in more complex areas. This is also more obvious for the CMXNeXt network, which also uses the segformer encoder. It fully proves the effectiveness of the segformer network for buildings. In the Vaihingen and USGS datasets, our model adeptly discerns small gaps between intricately clustered, dense buildings, ensuring accurate segmentation, and avoids problems such as tree shading and shadows to a certain extent. The Potsdam dataset is dominated by continuous large buildings. In the results, our method does not have any large-area omission and misdetection problems. The boundaries of the buildings are very clear. We have accurately delineated the small nonbuilding areas within large structures, with only a minimal number of detection errors.

2) Ablation Experiment: The quantitative experimental results of the three datasets are shown in Tables IV–VI. Among them, Our-rgb represents the extraction of only a single HRI; Our-sum and Our-cat represent the removal of only the original model CMFF module to be replaced by direct sum or cat; Our-RD represents the removal of the MSUD module, and all of them are replaced by the classical RD. Separate experiments were not conducted because the single DSM data contained less information to extract buildings effectively.

Comparative analysis of the experiment results reveals a significant improvement after fusion compared to the original single HRI data. The three datasets exhibited enhancements of approximately 0.5%, 1%, and 1.9%, respectively, underscoring the efficacy of DSM data. Figs. 8–10 show our prediction results for the three datasets. In the Vaihingen dataset, we choose one of the more exotic buildings. As shown in Fig. 8, at the heart of the dense structures lies a nonbuilding area. However, buildings in the periphery are much taller, thus causing the model to misclassify some areas as nonbuilding categories. During the extraction of HRI data, there are significant omissions, and the boundary details of shaded areas are not accurately delineated. The introduction of DSM data partially rectified inaccuracies in the shaded area boundaries, alleviating significant omissions in the extraction process. For the Potsdam dataset, we choose a large dense building as an example. Most of the small buildings inside are in the shaded area due to the influence of the surrounding tall buildings, which leads to greater difficulties in extraction. By DSM data, we are able to discriminate well between the building categories that are located in the internal shadowed areas, further improving the extraction of the buildings. This same point is evident in the USGS dataset. In the USGS dataset,

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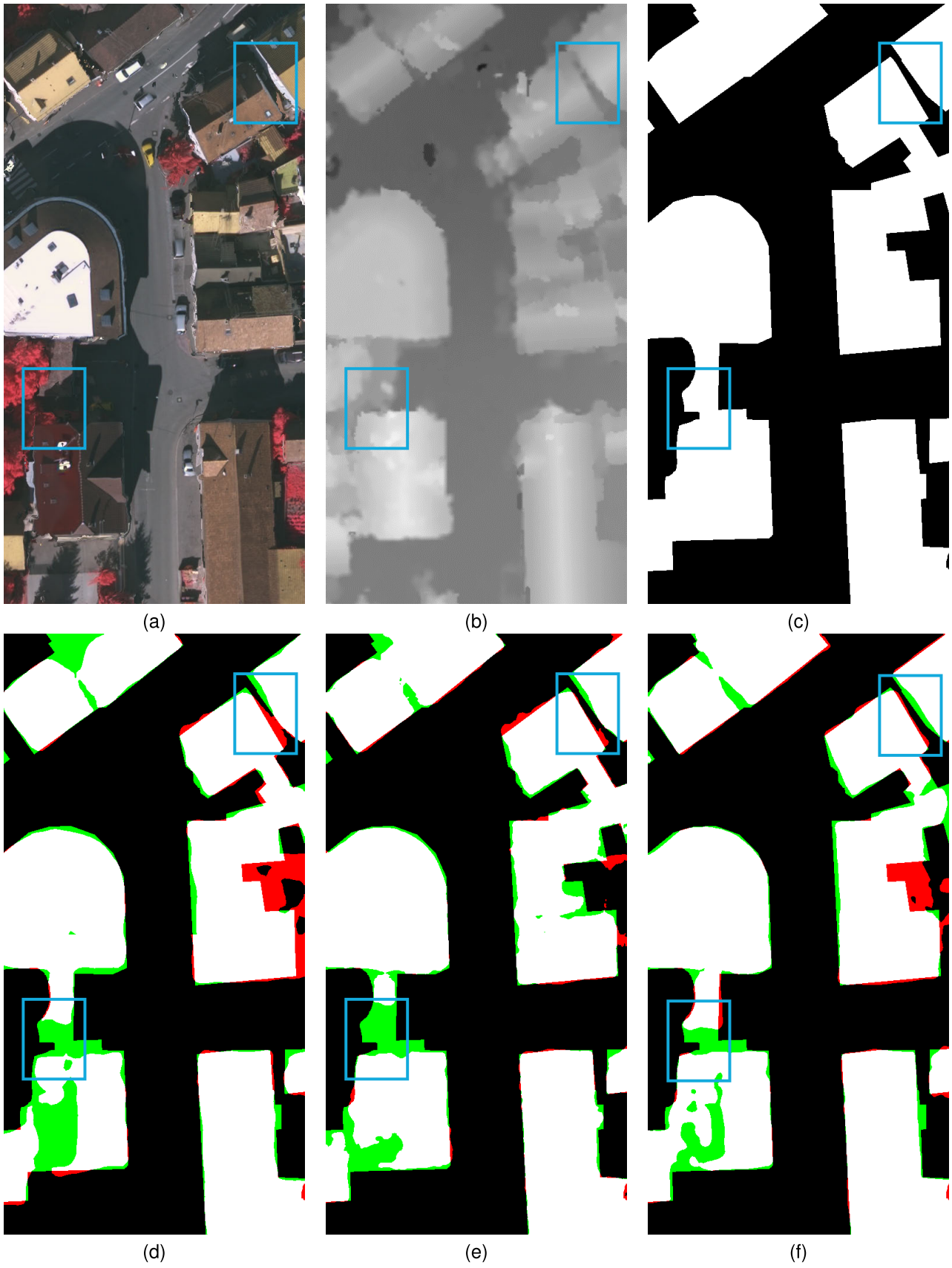


Fig. 5. Vaihingen comparison of the proposed method and other eight models. (a) HRI. (b) LiDAR. (c) GT. (d) CF-Net. (e) HAFNetE. (f) MultiModNet.

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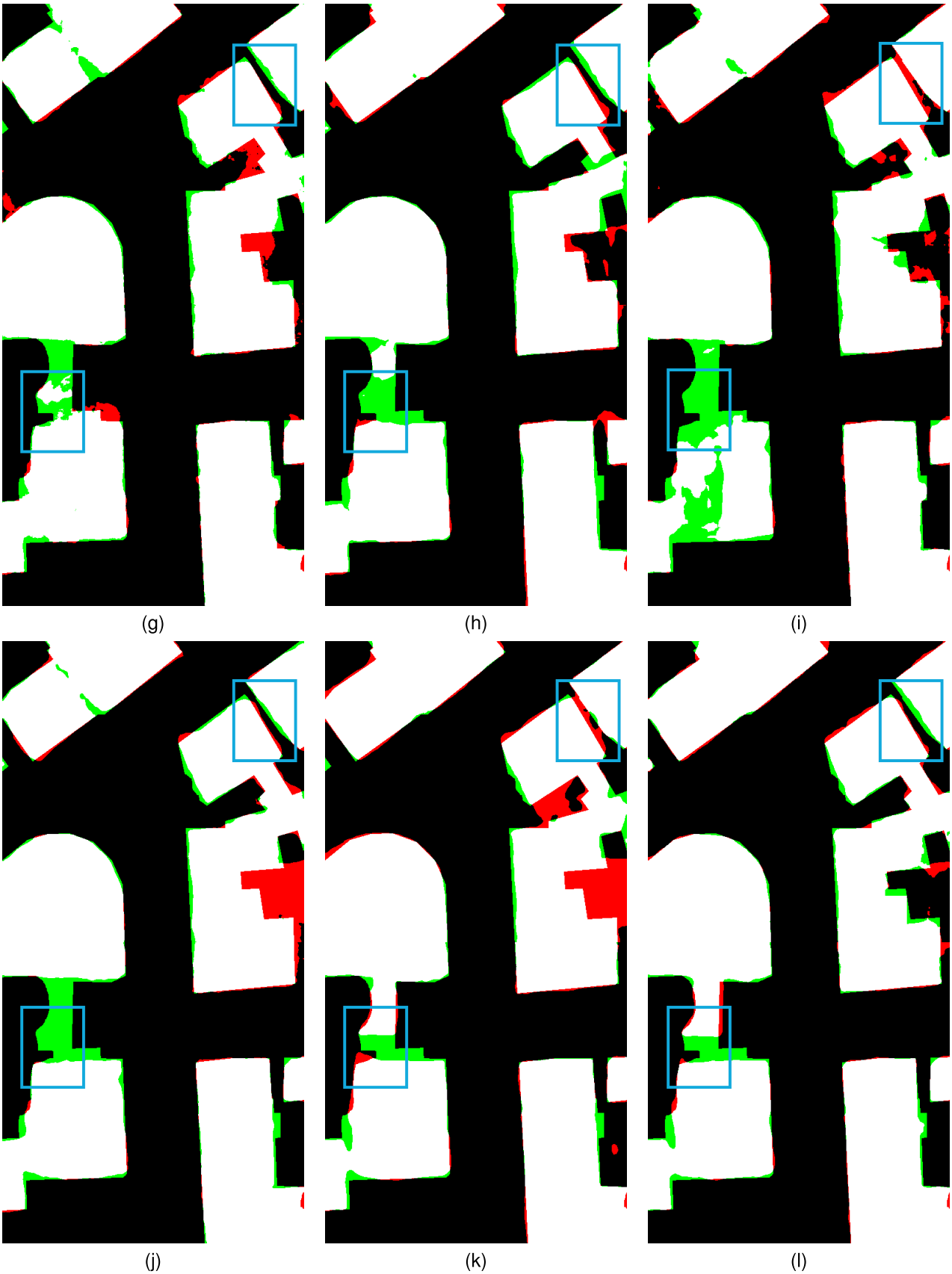


Fig. 5. *(Continued.)* Vaihingen comparison of the proposed method and other eight models. (g) CMFNet. (h) CMGFNet. (i) MGCNet. (j) SFAF-MA. (k) CMXNeXt. (l) Our. TP, FP, and FN are marked in white, red, and green, respectively.

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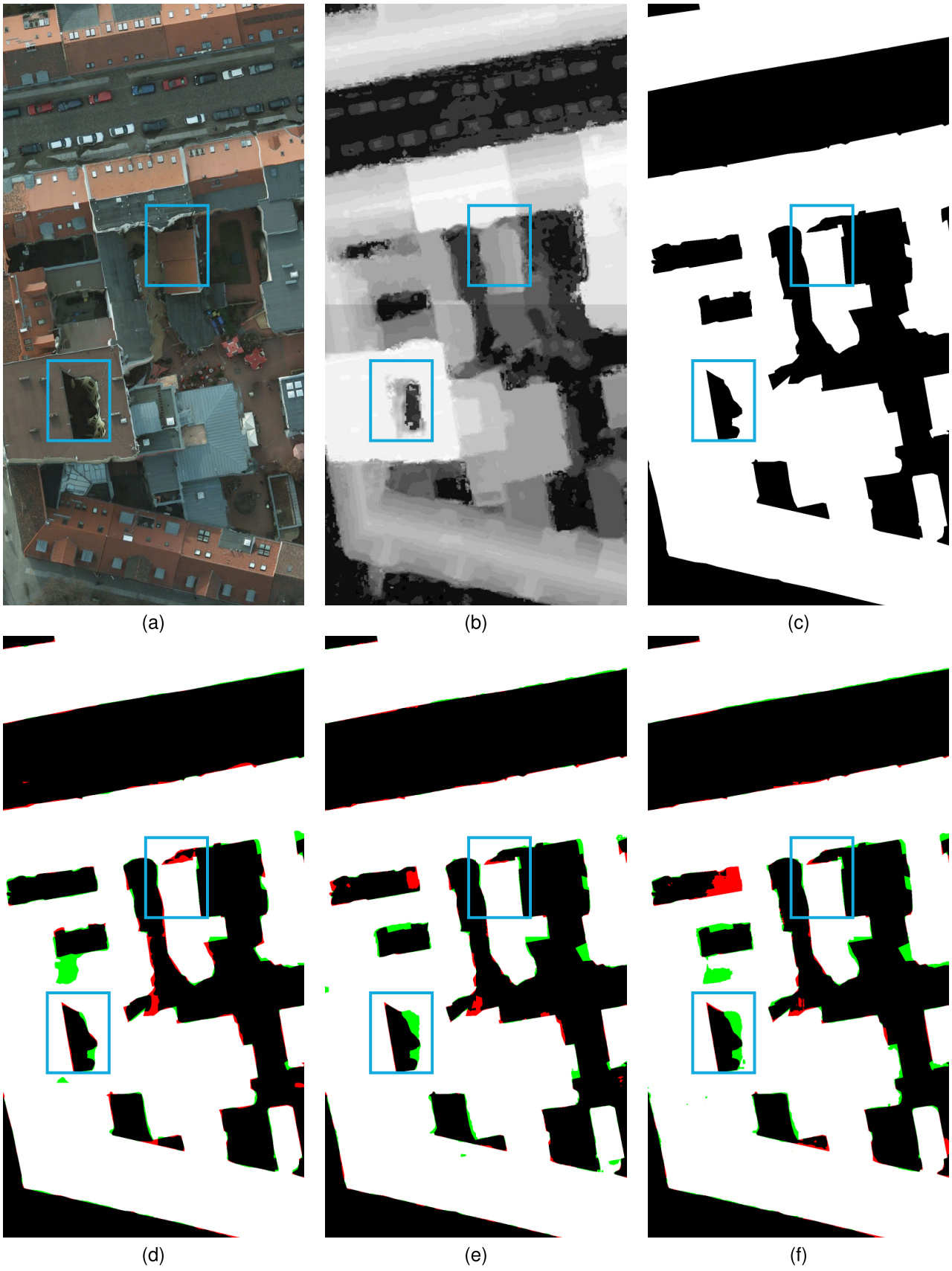


Fig. 6. Potsdam comparison of the proposed method and other eight models. (a) HRI. (b) LiDAR. (c) GT. (d) CF-Net. (e) HAFNetE. (f) MultiModNet.

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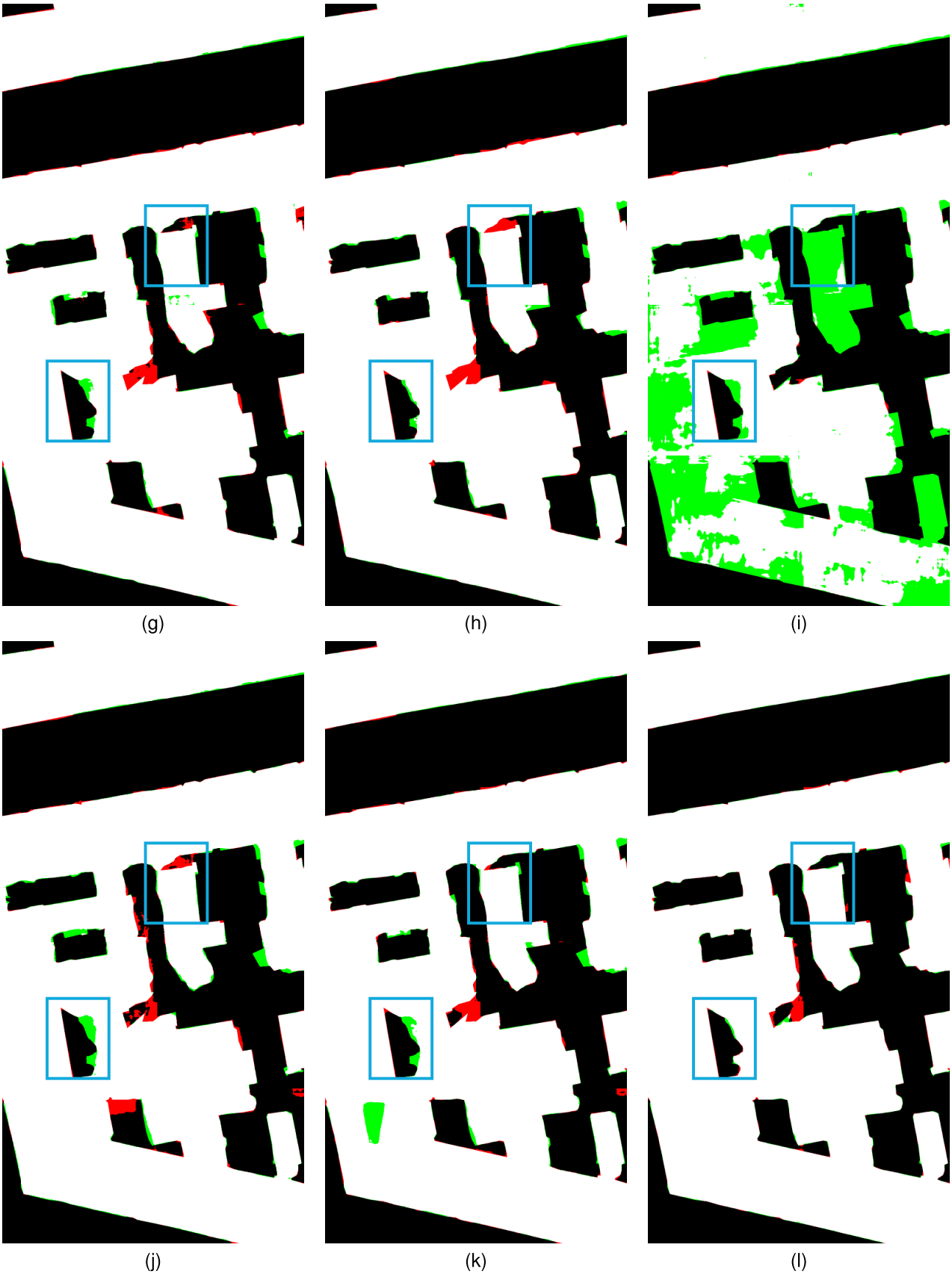


Fig. 6. *(Continued.)* Potsdam comparison of the proposed method and other eight models. (g) CMFNet. (h) CMGFNet. (i) MGCNet. (j) SFAF-MA. (k) CMXNeXt. (l) Our. TP, FP, and FN are marked in white, red, and green, respectively.

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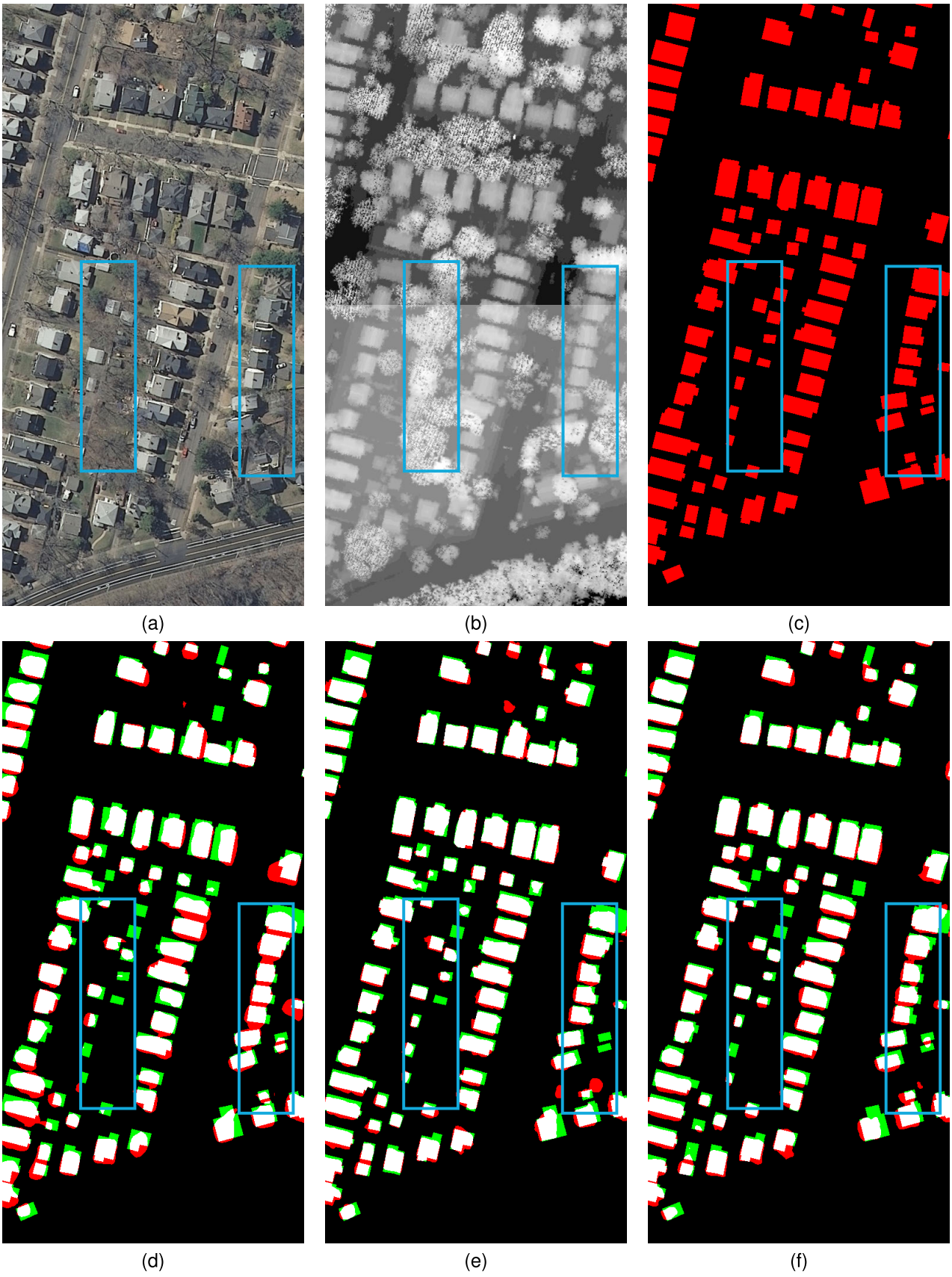


Fig. 7. USGS comparison of the proposed method and other eight models. (a) HRI. (b) LiDAR. (c) GT. (d) CF-Net. (e) HAFNetE. (f) MultiModNet.

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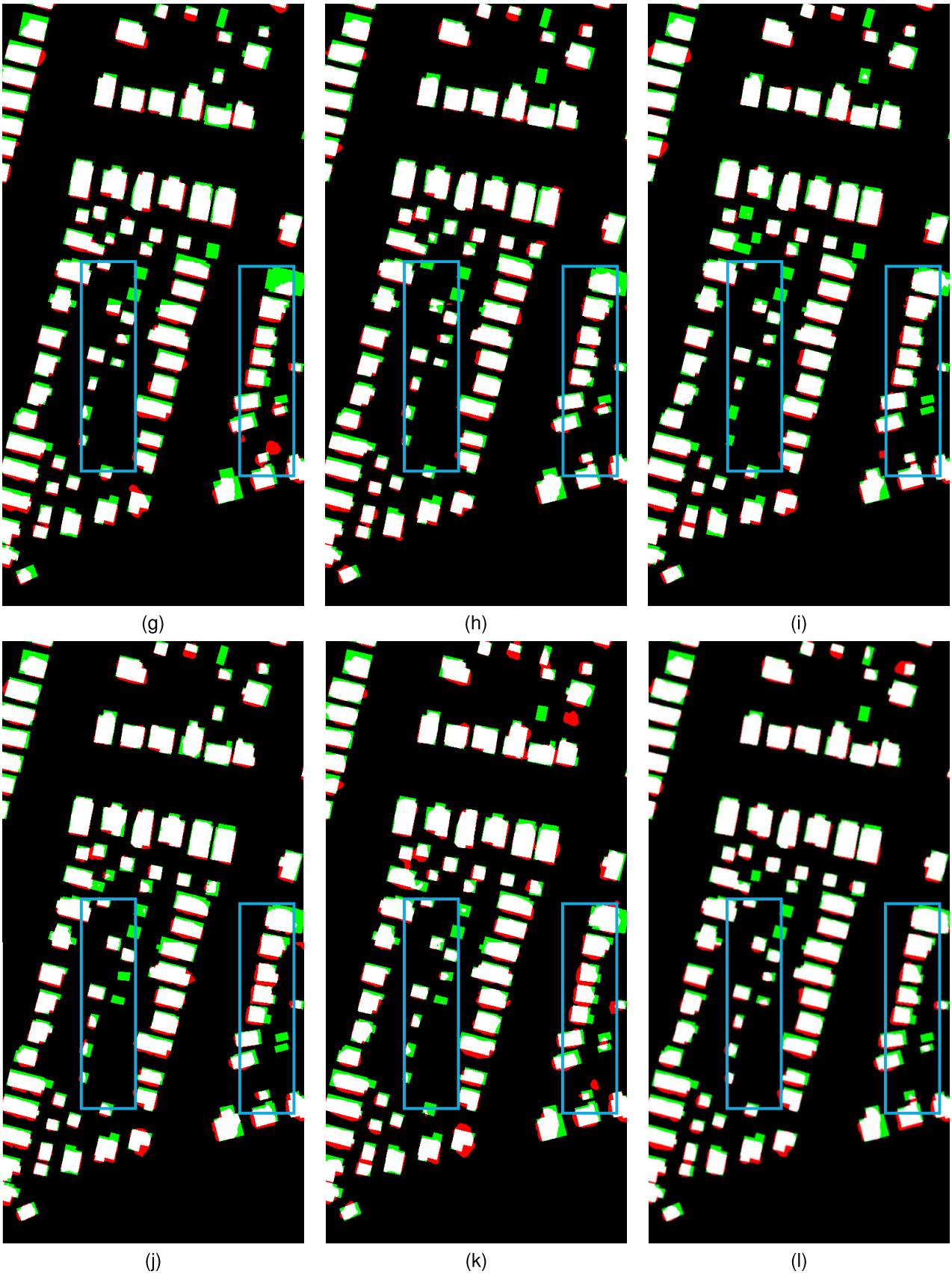


Fig. 7. *(Continued.)* USGS comparison of the proposed method and other eight models. (g) CMFNet. (h) CMGFNet. (i) MGCNet. (j) SFAF-MA. (k) CMXNeXt. (l) Our. TP, FP, and FN are marked in white, red, and green, respectively.

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TABLE IV
VAIHINGEN EVALUATION METRICS SCORE (%) OF ABLATION EXPERIMENT

Model-Name	CMFF	MSUD	ACC	F1-score	IoU
Our-rgb		✓	97.49	95.42	91.22
Our-sum		✓	97.55	95.56	91.50
Our-cat		✓	97.57	95.60	91.57
Our-RD	✓		97.51	95.51	91.40
Our	✓	✓	**97.64**	**95.72**	**91.80**

TABLE V
POTSDAM EVALUATION METRICS SCORE (%) OF ABLATION EXPERIMENT

Model-Name	CMFF	MSUD	ACC	F1-score	IoU
Our-rgb		✓	97.88	95.83	91.98
Our-sum		✓	98.04	96.16	95.59
Our-cat		✓	98.03	96.14	92.56
Our-RD	✓		98.06	96.19	92.67
Our	✓	✓	**98.16**	**96.39**	**93.04**

TABLE VI
USGS EVALUATION METRICS SCORE (%) OF ABLATION EXPERIMENT

Model-Name	CMFF	MSUD	ACC	F1-score	IoU
Our-rgb		✓	96.67	90.52	82.69
Our-sum		✓	96.78	90.94	83.38
Our-cat		✓	96.77	90.90	83.32
Our-RD	✓		96.85	91.11	83.67
Our	✓	✓	**97.03**	**91.65**	**84.59**

we select an area with more trees. It is evident that the addition of DSM data enhances the clarity of building boundaries, particularly those difficult to distinguish in HRI due to shadow. Furthermore, the boundaries of continuous buildings are more pronounced, contributing to the improvement of small building identification. This effectively underscores the substantial advantages offered by LiDAR data. Both quantitative and qualitative experimental results from the three datasets demonstrate the usefulness of LiDAR data for building extraction.

In addition, to further validate the effectiveness of the MSUD module, we replace it with an RD module. Comparing the experimental results, it is observed that the accuracy of the three datasets experiences a certain level of decline. Notably, the USGS dataset registers an approximate degradation of 1%, providing evidence for the efficacy of our MSUD module.

3) Impact of Different Encoder Frameworks: In order to verify the effectiveness of the segformer encoder, we conduct an experiment comparing it with traditional CNN. We choose several kinds of the most commonly used hierarchical CNN for building extraction. We further validate the effect of different segformer frameworks on building extraction. Table VII shows our validation for different encoder networks, and Table VIII shows our validation for different segformer encoder networks. Due to the relatively simple scenes in the Vaihingen and Potsdam datasets, which result in a lower overall extraction difficulty and less distinct differences, we have chosen the more complex USGS dataset for conducting comparative experiments between different frameworks.

In the comparison of experimental results, we can find that the segformer network has a large improvement over CNN. Compared to the best DenseNet-121 network, there are also about 1.5% improvements. This result fully confirms the usefulness of the segformer encoder. For the different segformer networks, segformer-b4 shows the best results. Although segformer-b5 has more hidden layers, it does not show the effect prominently enough. Therefore, we finally decided to use segformer-b4 as the encoder framework.

V. DISCUSSION

The challenge of building extraction is the determination of boundaries and the diversity of building spectra. In specific regions, buildings are densely distributed, posing challenges for segmentation and often leading to issues such as blurred boundaries, omissions, or erroneous extractions. For the increasingly complex building categories and topographic conditions, there is a huge obstacle for single-source data to meet the needs of high-precision extraction. In addition, the traditional CNN has become relatively weak in the face of increasingly complex buildings. It is difficult to effectively and completely extract the features of buildings. Therefore, in this article, we propose to use a segformer network to break through the limitation of traditional CNN and use

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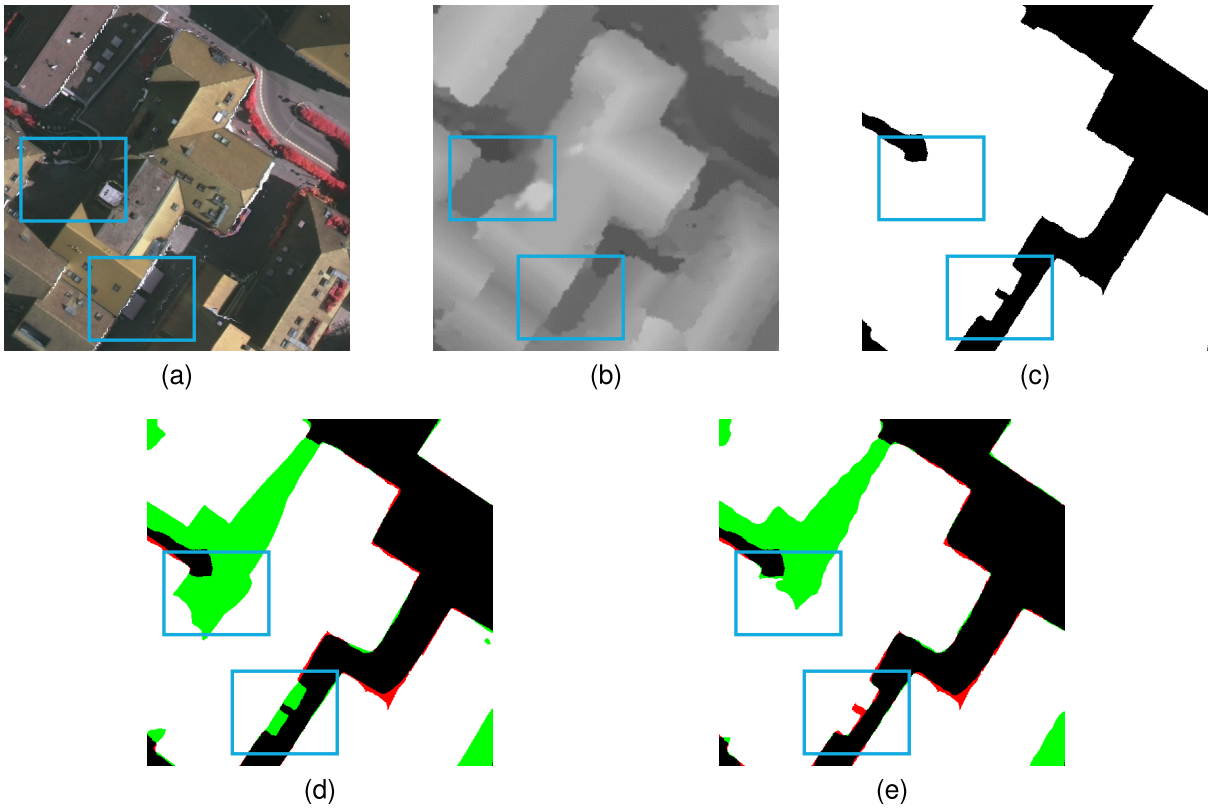


Fig. 8. Selected images from Vaihingen and the results of building extraction produced by different models. (a) HRI. (b) LiDAR. (c) Ground truth. (d) Our-rgb. (e) Our. TP, FP, and FN are marked in white, red, and green, respectively.

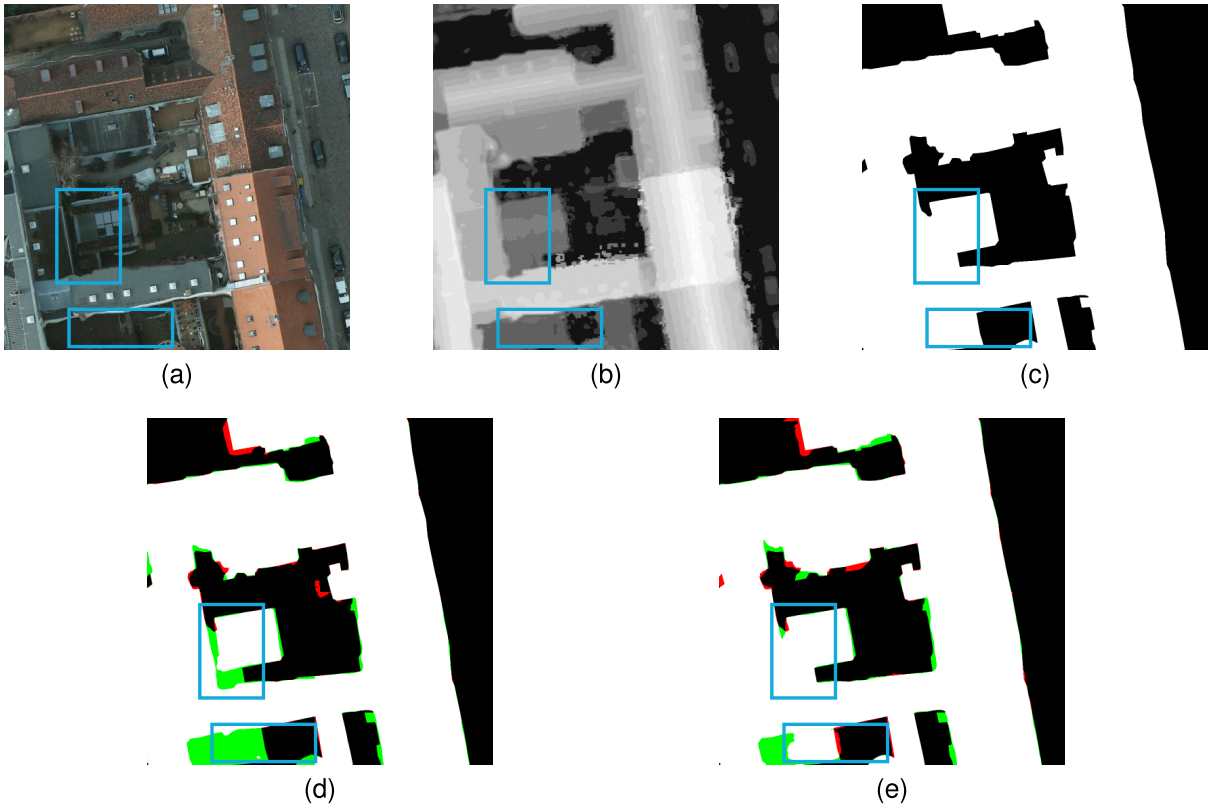


Fig. 9. Selected images from Potsdam and the results of building extraction produced by different models. (a) HRI. (b) LiDAR. (c) Ground truth. (d) Our-rgb. (e) Our. TP, FP, and FN are marked in white, red, and green, respectively.

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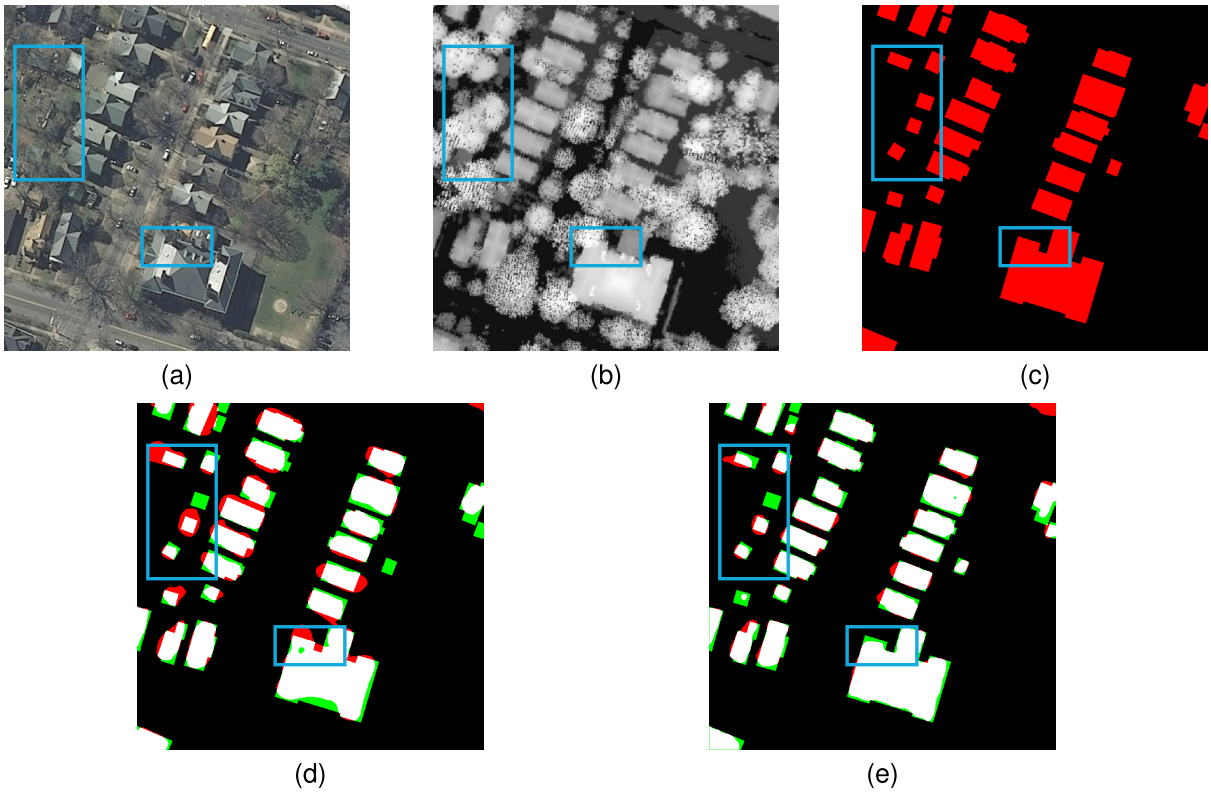


Fig. 10. Selected images from USGS and the results of building extraction produced by different models. (a) HRI. (b) LiDAR. (c) Ground truth. (d) Our-rgb. (e) Our. TP, FP, and FN are marked in white, red, and green, respectively.

TABLE VII
DIFFERENT ENCODER FRAMEWORK RESULTS

Encoder-Name(USGS)	OA(%)	F1-score(%)	IoU(%)	Parameter(M)
VGG-19	96.64	90.45	82.56	42.11
DenseNet-121	96.73	90.74	83.05	34.56
Efficient-b0	96.47	90.02	81.86	9.27
ResNet-34	96.63	90.58	82.78	45.08
Segformer-b4	**97.03**	**91.65**	**84.59**	**119.80**

TABLE VIII
DIFFERENT SEGFORMER FRAMEWORK RESULTS

Encoder-Name(USGS)	OA(%)	F1-score(%)	IoU(%)	Parameter(M)
Segformer-b0	96.71	90.68	82.95	7.53
Segformer-b1	96.81	91.11	83.67	33.50
Segformer-b2	96.89	91.31	84.02	54.57
Segformer-b3	96.93	91.40	84.18	92.48
Segformer-b4	**97.03**	**91.65**	**84.59**	119.80
Segformer-b5	96.96	91.42	84.20	**163.76**

global information to extract the features of buildings in multisource data. Later, we weigh the global position of the fused multisource data according to the self-attention mechanism to ensure the fusion of multisource data. We make full use of the advantages of two data sources to address the shortcomings of each other. Moreover, compared with CNN, our model has made a big breakthrough. Even though the same encoder is used, our model still achieves good results compared to other networks. In addition, it also has a great improvement compared with other models of transformer and segformer, which fully demonstrates the effectiveness of the individual modules. This affirmation is well supported by the comparison of experimental results across the three datasets.

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VI. Conclusion

The main objective of our work is to extract buildings from remote sensing images. We propose an end-to-end building extraction network for multisource fusion of HRIs and LiDAR data based on segformer. The network fully extracts the feature of building targets by exploiting the unique characteristics of the segformer framework. Subsequently, the extracted multisource features are fused using the CMFF module, resulting in complementary advantages. In the decoder part, high-level semantic information and low-level features are fused through the MSUD module. It ensures the full use of features at different scales.

In the experimental part, we validate and analyze the effectiveness of the model using two public datasets and one dataset processed by ourselves. Three different datasets contain a variety of complex terrain conditions, such as urban and suburban areas and rural areas, with complex and variable building types, which can well validate the performance of the model. The effectiveness of our proposed network is verified by comparing it with several other multisource fusion building extraction and semantic segmentation networks. Following this, the effectiveness of the proposed method is further analyzed using ablation experiments. However, the drawback of this network lies in the more complex network structure and the larger number of model parameters. Thus, in future work, we will consider using the distillation structure to further compress the parameters of the network and improve the efficiency of the model.

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